



AlphaAgent

Complete Technical Reference Manual

A production-grade, multi-agent quantitative trading intelligence system.
9 specialist agents · 163 factors · 11 quant engine modules · LangGraph orchestration

Version 6.1 — Phase 6 Complete

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Platform: Windows 11 · Python 3.11 · FastAPI · LangGraph

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Chapter 1 — System Overview & Philosophy

AlphaAgent is a multi-agent quantitative trading signal system that combines nine independent analytical perspectives into a single probability-scored trading recommendation. Every signal is produced through a rigorous, mathematically grounded pipeline with full transparency: every factor is scored 0–100, every agent reports its reasoning, and the orchestrator exposes the Bayesian posterior probability, Shannon entropy, and position-sizing multiplier to the end user.

Design Principles

- **No black boxes.** Every factor that moves a probability is named, scored, and shown in the dashboard with its raw value and interpretation.
- **Correlation penalty.** Agents that share data sources are penalized in the Bayesian fusion step so that correlated evidence is not double-counted.
- **Hard circuit breakers.** The Risk Agent can override all other agents with a HALT command (position size = 0) when Black Swan, Flash Crash, or Hawkes cascade signals fire.
- **Dynamic confidence.** Each agent passes a `confidence` weight (0–1) that scales its evidence in the Bayesian update; low-confidence agents contribute less.
- **Time-horizon awareness.** The final probability blend is reweighted by a horizon multiplier map (1D→technical-dominant, 1Y→fundamental-dominant) before delivery to the user.

Signal Pipeline Summary

Step 1: MarketData fetches OHLCV, fundamentals, options, news, FRED macro in parallel (cached).

Step 2: 9 agents run in parallel via ThreadPoolExecutor. Each returns $P(\text{up}) \in [0,1]$ + factor scores.

Step 3: Bayesian fusion combines agent probabilities with correlation penalties and confidence weights.

Step 4: Risk circuit breaker applies HALT / size-reduction overrides.

Step 5: HMM regime detection + signal decay half-life computed.

Step 6: Horizon reweighting applied per user-selected time frame.

Step 7: Final SignalPacket returned: direction, $P(\text{up})$, conviction %, entropy, size multiplier.

Factor Count by Agent

Agent	Factors	Data Sources	Primary Theory
Technical	30	yfinance OHLCV, options chain	Price action, options microstructure, TDA
Fundamental	28	yfinance financials, FRED	DCF, Fama-French 5-Factor, CAPM
Macro	26	FRED API, yfinance ETFs	Yield curve, monetary policy, global breadth
Sentiment	18	NewsAPI, yfinance news, ChromaDB	NLP RAG, social sentiment, options flow
Insider	14	SEC EDGAR, yfinance, House/Senate Watcher	Microstructure, Kyle's Lambda, clustering
Risk	16	yfinance OHLCV + options	EVT, GARCH, Hawkes, Quasi-MC, KL divergence
Geopolitical	15	FRED, yfinance commodities/ETFs	GPR index, Caldara-Iacoviello, oil signals
Volatility	12	yfinance OHLCV + options	GARCH, Kalman, IV term structure
Currency	12	yfinance FX pairs, FRED rates	Carry trade, DXY regime, EM stress
Total	171		

Chapter 2 — Architecture Deep Dive

Execution Graph (LangGraph)

```

START
|
▼
[data_ingestion_node]
  Pre-warms: OHLCV(1y), financials, info, news
  |
  ▼
[run_agents_node] ← ThreadPoolExecutor(max_workers=9)
  ├── TechnicalAgent.analyze()
  ├── FundamentalAgent.analyze()
  ├── MacroAgent.analyze()
  ├── SentimentAgent.analyze()
  ├── InsiderAgent.analyze()
  ├── RiskAgent.analyze()
  ├── GeopoliticalAgent.analyze()
  ├── VolatilityAgent.analyze()
  └── CurrencyAgent.analyze()
  |
  ▼
[portfolio_manager_node]
  Risk circuit breaker check
  BayesianFusion.update() × 8 agents (risk excluded)
  HMM regime detection
  SignalDecay half-life
  Build SignalPacket
  |
  ▼
END → HTTP Response

```

Module Dependencies

Module	Imports From	Purpose
<code>api/main.py</code>	orchestrator, agents, database	FastAPI HTTP layer, horizon weighting
<code>orchestrator/graph.py</code>	agents.*, quant_engine.bayesian, quant_engine.hmm	LangGraph workflow compilation
<code>agents/*.py</code>	data.market, quant_engine.*, config.settings_manager	9 individual analysis agents
<code>quant_engine/*.py</code>	scipy, numpy, arch, hmmlearn, ripser	Mathematical computation kernels
<code>data/market.py</code>	yfinance, fredapi, chromadb	Cached data access layer
<code>database/manager.py</code>	sqlalchemy, sqlite3	Trade/signal persistence
<code>frontend/index.html</code>	(self-contained)	Single-file dashboard

Chapter 3 — Data Layer (MarketData)

The `MarketData` class in `data/market.py` is the single data access point for all agents. It wraps `yfinance` and `fredapi` with in-memory TTL caching to prevent redundant API calls during a single analysis run.

Key Methods

Method	Returns	Cache TTL	Source
<code>get_ohlcv(period)</code>	DataFrame: Open/High/Low/Close/Volume	300s	yfinance
<code>get_returns(period)</code>	Series: log returns	300s	yfinance (derived)
<code>get_financials()</code>	Dict: income/balance/cashflow	604800s (1w)	yfinance
<code>get_info()</code>	Dict: ticker metadata	86400s (1d)	yfinance
<code>get_options()</code>	Dict: calls/puts DataFrames per expiry	300s	yfinance options chain
<code>get_news()</code>	List[Dict]: recent headlines	1800s	yfinance + NewsAPI
<code>get_fred(series_id)</code>	Series: macro time series	3600s	FRED API (fredapi)

Data Pre-warming

The `data_ingestion_node` calls `get_ohlcv("1y")`, `get_financials()`, and `get_info()` before agents start. This ensures all agents share one API call instead of each making their own, reducing latency from ~90s to ~30–60s.

yfinance Multi-Index Fix

Critical Bug (fixed): yfinance ≥ 0.2 returns a MultiIndex column DataFrame when downloading with `group_by="ticker"`. All agents use `df.columns = df.columns.get_level_values(0)` to flatten the MultiIndex before accessing columns like `"Close"`.

Chapter 4 — Agent Framework & State Schema

Core Data Classes (agents/state.py)

FactorScore

```
@dataclass
class FactorScore:
    name: str          # Display name, e.g. "RSI (14)"
    value: float        # Raw computed value (e.g. 72.3)
    score: float        # Normalized 0-100 (50 = neutral, 100 = maximum bullish)
    interpretation: str # Plain-English explanation
```

The **score** convention: values near 0 are bearish, values near 100 are bullish, 50 is neutral. The composite agent probability is derived by averaging all factor scores and mapping $[0,100] \rightarrow [0,1]$.

AgentResult

```
@dataclass
class AgentResult:
    agent_name: str          # Unique agent identifier
    vote: Direction          # LONG | SHORT | HOLD
    probability_up: float     # P(price up) ∈ [0.0, 1.0]
    confidence: float        # Confidence in this estimate ∈ [0.0, 1.0]
    reasoning: str           # Narrative explanation
    factor_scores: dict[str, FactorScore]
    warnings: list[str]
    computation_time_ms: float
    timestamp: datetime
```

SignalPacket

```
@dataclass
class SignalPacket:
    ticker: str
    direction: Direction          # Final vote
    conviction_pct: float         # |P(up) - 0.5| × 2 × 100
    confidence: Confidence        # HIGH/MEDIUM/LOW enum
    agent_results: list[AgentResult] # All 9 agent outputs
    regime: RegimeResult         # HMM regime
    holding_period: HoldingPeriod # Signal decay half-life
    override_active: bool
    override_reason: str
    warnings: list[str]
```

HoldingPeriod

```
@dataclass
class HoldingPeriod:
    half_life_days: float        # Signal half-life (Ornstein-Uhlenbeck mean reversion)
    optimal_hold_min: int        # Recommended minimum hold (days)
    optimal_hold_max: int        # Recommended maximum hold (days)
    signal_strength: float       # [0,1] – affects recommended size
```

Direction Enums

Enum	Values
Direction	LONG, SHORT, HOLD, NEUTRAL
Confidence	HIGH (conviction>0.4), MEDIUM (>0.2), LOW
Regime	BULL, BEAR, CRISIS, SIDEWAYS
VolRegime	LOW, NORMAL, HIGH, EXTREME

BaseAgent Interface

```
class BaseAgent(ABC):
    name: str # Must be set in subclass

    def run(self, ticker: str, market_data: MarketData) -> AgentResult:
        """Public entry point. Times execution, catches all exceptions."""
        t0 = time.time()
        try:
            result = self.analyze(ticker, market_data)
            result.agent_name = self.name
            result.computation_time_ms = (time.time() - t0) * 1000
            return result
        except Exception as e:
            return AgentResult(agent_name=self.name, vote=Direction.HOLD,
                               probability_up=0.5, confidence=0.0,
                               reasoning=f"AGENT FAILURE: {e}")

    @abstractmethod
    def analyze(self, ticker: str, market_data: MarketData) -> AgentResult:
        """Override this to implement the agent logic."""
```

Chapter 5 — LangGraph Orchestrator

Node 1: data_ingestion_node

Pre-warms the MarketData cache. Calls `get_ohlcv("1y")`, `get_financials()`, `get_info()`, and fetches news. Because all results are cached in-process, subsequent agent calls to these methods return instantly from cache.

Node 2: run_agents_node

Uses Python's `ThreadPoolExecutor` with up to 9 workers to run all registered agents in parallel. Each agent gets its own thread. Results are collected via `as_completed()` futures. If any agent raises an exception, it is caught and replaced with a neutral `AgentResult(probability_up=0.5)`.

Node 3: portfolio_manager_node

This is the aggregation brain. It performs six sequential operations:

1. **Risk circuit breaker check** — reads Risk Agent reasoning text for keywords: `CRITICAL RISK`, `BLACK SWAN`, `FLASH CRASH` → sets `multiplier = 0.0`, `direction = HOLD`. `HIGH RISK` → `multiplier = 0.5`. `GEO_SHOCK` → `multiplier ≤ 0.35`.
2. **Bayesian fusion** — feeds 8 agent probabilities (risk excluded) into `BayesianFusion` with correlation penalties. See Chapter 6.
3. **Direction assignment** — `P(up) > 0.55 → LONG`, `P(up) < 0.45 → SHORT`, else `HOLD`.
4. **Conviction calculation** — `conviction = |P(up) - 0.5| × 2` (range: 0=neutral to 1=max).
5. **HMM regime detection** — fits 3-state Gaussian HMM to 1-year returns. Labels current market as BULL/BEAR/CRISIS.
6. **Signal decay** — estimates holding period via Ornstein-Uhlenbeck half-life of price reversion.

Correlation Map (double-counting prevention)

Agent	Correlation Penalty	Reason
sentiment	0.20	Partly overlaps with news used by macro
volatility	0.40	Highly overlaps with risk agent's GARCH
geopolitical	0.25	Partly overlaps with macro's global breadth
currency	0.20	Partly overlaps with macro's DXY regime
technical, fundamental, macro, insider	0.00	Independent data sources

Chapter 6 — Bayesian Fusion Engine

Source: `quant_engine/bayesian.py`

Theory

Standard probability averaging ignores evidence strength and correlation. Bayesian Fusion operates in **log-odds space** where each agent's probability is converted to a log-odds (logit) and the contributions are summed — this is the correct Bayesian sequential update rule for independent evidence.

Mathematical Formulation

Initialization:

$$\text{LogOdds}_0 = \ln\left(\frac{\pi_0}{1 - \pi_0}\right)$$

where $\pi_0 = 0.5$ (uniform prior: no directional bias before evidence).

Evidence from agent i :

$$E_i = \ln\left(\frac{p_i}{1 - p_i}\right) \cdot c_i \cdot (1 - \rho_i)$$

where p_i = agent probability, c_i = confidence weight, ρ_i = correlation penalty.

Sequential update:

$$\text{LogOdds}_k = \text{LogOdds}_{k-1} + E_k$$

Posterior conversion (sigmoid):

$$P(\text{up}) = \frac{1}{1 + e^{-\text{LogOdds}_{\text{final}}}}$$

Shannon Entropy

$$H = -[p \log_2 p + (1 - p) \log_2 (1 - p)]$$

- $H = 0.0$: Perfect certainty ($p = 0$ or $p = 1$)
- $H = 1.0$: Total confusion ($p = 0.5$)
- AlphaAgent warns when $H > 0.9$: agents strongly disagree

Worked Example

Suppose technical agent reports $p=0.72$ (bullish) with confidence= 0.9 , no correlation.
 Macro agent reports $p=0.48$ (slightly bearish) with confidence= 0.7 , correlation= 0.0 .
 Sentiment reports $p=0.65$ with confidence= 0.6 , correlation= 0.20 .

```
Prior log-odds:      ln(0.5/0.5)      = 0.000
Technical:          ln(0.72/0.28) × 0.9 × 1.0 = 0.942 × 0.9 = +0.848
Macro:              ln(0.48/0.52) × 0.7 × 1.0 = -0.167 × 0.7 = -0.117
Sentiment:          ln(0.65/0.35) × 0.6 × 0.8 = 0.619 × 0.48 = +0.297
Final log-odds:      0.000 + 0.848 - 0.117 + 0.297 = 1.028
Posterior P(up):     1 / (1 + exp(-1.028)) = 0.736 → LONG
```

Chapter 7 — GARCH Volatility Model

Source: `quant_engine/garch.py` · Library: `arch`

Theory

GARCH (Generalized Autoregressive Conditional Heteroskedasticity) models the fact that financial volatility clusters — high-volatility days tend to follow high-volatility days. Unlike historical volatility which assumes constant vol, GARCH forecasts *tomorrow's* volatility based on recent shocks and the persistence of variance.

GARCH(1,1) Equation

Return equation:

$$r_t = \mu + \epsilon_t, \quad \epsilon_t = \sigma_t z_t, \quad z_t \sim N(0, 1)$$

Variance equation:

$$\sigma_t^2 = \omega + \alpha \epsilon_{t-1}^2 + \beta \sigma_{t-1}^2$$

Parameters:

- ω (omega): Long-run variance constant
- α (alpha[1]): News impact — how much last period's shock affects today's variance
- β (beta[1]): Persistence — how much last period's variance persists today

Persistence: $\alpha + \beta < 1$ for stationarity. Close to 1 = long memory.

Long-run (unconditional) volatility:

$$\sigma_\infty = \sqrt{\frac{\omega}{1 - \alpha - \beta}}$$

Annualized from daily:

$$\sigma_{\text{ann}} = \sigma_{\text{daily}} \times \sqrt{252} \times 100$$

Multi-Step Forecast

For horizon h , GARCH variance forecast converges to the long-run variance:

$$\hat{\sigma}_{t+h}^2 = \sigma_{\infty}^2 + (\alpha + \beta)^{h-1}(\sigma_{t+1}^2 - \sigma_{\infty}^2)$$

5-day aggregated volatility:

$$\sigma_{5d} = \sqrt{\sum_{j=1}^5 \hat{\sigma}_{t+j}^2}$$

Regime Classification

Regime	Vol Percentile	Meaning
LOW	< 25th	Calm market, signals more reliable
NORMAL	25th–75th	Average vol environment
HIGH	75th–95th	Elevated volatility, wider stops needed
EXTREME	> 95th	Crisis-level vol, reduce position size

GARCHResult Fields

Field	Type	Description
<code>vol_1day</code>	float	1-day ahead annualized vol forecast (%)
<code>vol_5day</code>	float	5-day aggregated annualized vol (%)
<code>vol_10day</code>	float	10-day aggregated annualized vol (%)
<code>current_vol</code>	float	Most recent realized conditional vol (%)
<code>vol_percentile</code>	float	Percentile rank of current vol vs history
<code>vol_regime</code>	str	LOW / NORMAL / HIGH / EXTREME
<code>alpha</code>	float	News impact coefficient
<code>beta</code>	float	Variance persistence coefficient
<code>persistence</code>	float	$\alpha + \beta$ (must be < 1)
<code>aic</code> / <code>bic</code>	float	Model information criteria (lower=better)

Implementation Note

Returns are scaled by 100 (percentage) before fitting because the `arch` library is numerically more stable when the scale of returns is $\sim 1\text{--}5$ rather than $0.01\text{--}0.05$. All outputs are divided back by the scale factor before reporting.

Chapter 8 — Hidden Markov Model (Regime Detection)

Source: `quant_engine/hmm.py` · Library: `hmmlearn`

Theory

Markets cycle through unobservable (hidden) states — Bull, Bear, Crisis — which manifest as different return distributions. A Gaussian HMM learns these states unsupervised from return data: each state has a characteristic mean return and volatility. The model then assigns a probability to each state for the current period.

Model Specification

3-state Gaussian HMM:

$$P(r_t | S_t = k) = \mathcal{N}(\mu_k, \sigma_k^2)$$

Transition matrix A :

$$A_{ij} = P(S_t = j | S_{t-1} = i)$$

State assignment: States are sorted by conditional volatility:

- State 0 (lowest σ) → BULL regime
- State 1 (medium σ) → BEAR regime
- State 2 (highest σ) → CRISIS regime

Posterior probabilities via Forward-Backward algorithm:

$$\gamma_t(k) = P(S_t = k | r_1, \dots, r_T)$$

RegimeResult Fields

Field	Type	Description
<code>current_regime</code>	str	BULL / BEAR / CRISIS
<code>probabilities</code>	dict	{bull: 0.72, bear: 0.21, crisis: 0.07}
<code>transition_risk</code>	dict	{to_bear: 0.15, to_crisis: 0.03}
<code>state_means</code>	list[float]	Daily return mean per state (sorted by vol)
<code>state_vols</code>	list[float]	Daily return std per state (sorted)
<code>regime_duration_days</code>	int	Consecutive days in current regime

Usage in Orchestrator

The HMM is run in `portfolio_manager_node` on 1-year of daily returns. The resulting `RegimeResult` is stored in the `SignalPacket.regime` field and displayed in the dashboard's regime card. It does *not* modify the probability directly — it provides regime context for the analyst.

Chapter 9 — Monte Carlo Simulation (GBM)

Source: `quant_engine/monte_carlo.py`

Geometric Brownian Motion

Continuous-time SDE:

$$dS = \mu S dt + \sigma S dW_t$$

Discretized Euler-Maruyama (Itô form):

$$S_{t+1} = S_t \cdot \exp \left[\left(\mu - \frac{\sigma^2}{2} \right) \Delta t + \sigma \sqrt{\Delta t} Z_t \right]$$

where $Z_t \sim N(0, 1)$, $\Delta t = 1$ (daily), μ = daily drift, σ = daily volatility. **Why the $-\sigma^2/2$ adjustment (Itô correction)?** Without it, $E[S_t]$ would be biased upward. The adjustment ensures $E[S_T] = S_0 e^{\mu T}$.

Implementation (Vectorized)

```
# Generate all shocks at once: shape (days, num_paths)
Z = np.random.standard_normal((days, num_paths))
drift_adj = drift_daily - 0.5 * vol_daily**2
daily_returns = np.exp(drift_adj + vol_daily * Z)
cum_returns = np.cumprod(daily_returns, axis=0) # Cumulative product per path
final_prices = current_price * cum_returns[-1, :] # Last day x num_paths
```

GARCH-Dynamic Version

Instead of constant vol, each day t uses GARCH-forecasted volatility $\hat{\sigma}_t$:

```
for t in range(1, days + 1):
    vol_t = garch_forecasts[t-1] # Day-specific GARCH forecast
    drift_adj = drift_daily - 0.5 * vol_t**2
    paths[t] = paths[t-1] * np.exp(drift_adj + vol_t * Z[t-1])
```

Confidence Intervals

CI Level	Percentiles Used	Interpretation
68% (1 σ)	[16th, 84th]	1 standard deviation range
95% (2 σ)	[2.5th, 97.5th]	Standard 95% confidence band
99% (3 σ)	[0.5th, 99.5th]	Extreme scenario range

MonteCarloResult Fields

Field	Description
<code>prob_above_current</code>	% of 10,000 paths ending above entry price
<code>expected_return_pct</code>	Mean return across all paths (%)
<code>ci_95</code>	ConfidenceInterval(low=..., high=...) in dollars
<code>max_simulated_price</code>	Worst-case tail upside across all paths
<code>min_simulated_price</code>	Worst-case tail downside across all paths

Chapter 10 — Quasi-Monte Carlo (Sobol Sequences)

Source: `quant_engine/quasi_mc.py` · Library: `scipy.stats.qmc.Sobol`

Why QMC?

Standard pseudorandom Monte Carlo converges at the *probabilistic* rate $O(N^{-1/2})$. Quasi-random sequences (Sobol, Halton) use a structured low-discrepancy design that fills the sample space more uniformly, achieving the *deterministic* rate $O((\log N)^d/N)$. In practice this means 10–100× tighter confidence intervals for the same number of paths.

Sobol Sequence Construction

Sobol sequences are based on **Van der Corput sequences** in base 2, extended to multi-dimensional spaces via direction numbers (Joe & Kuo 2003). Each dimension is a binary reflected Gray code sequence that minimizes the *star discrepancy*:

$$D_N^* = \sup_{[0,1]^d} \left| \frac{\#\{x_i \in B\}}{N} - \text{Vol}(B) \right|$$

Implementation: `scipy.stats.qmc.Sobol(d=days, scramble=True)` generates an $(N \times d)$ uniform matrix which is then transformed via the inverse normal CDF to produce quasi-normal shocks:

$$Z_{ij} = \Phi^{-1}(u_{ij}), \quad u_{ij} \in \text{Sobol}(N, d)$$

Efficiency Comparison

Method	Convergence Rate	Effective Sample Size	Tighter CIs?
Pseudorandom MC	$O(N^{-1/2})$	N	Baseline
Sobol QMC	$O((\log N)^d/N)$	~10N (empirical)	Yes, ~3× narrower

Power-of-2 Requirement

Sobol sequences achieve maximum uniformity guarantees when the number of paths N is a power of 2. AlphaAgent automatically rounds up: `_round_to_power2(n)`. Default: 4096 paths ($= 2^{12}$).

Chapter 11 — Extreme Value Theory (EVT)

Source: `quant_engine/evt.py` · Libraries: `scipy.stats.genpareto`, `scipy.stats.genextreme`

Overview

Standard VaR assumes returns are normally distributed — but financial returns have fat tails. EVT focuses exclusively on the tail of the distribution using two rigorous approaches: Peaks-Over-Threshold (POT) with a Generalized Pareto Distribution, and Block Maxima with a Generalized Extreme Value distribution.

Method 1: POT — Generalized Pareto Distribution

Automatic threshold selection (Mean Excess Function): The optimal threshold u^* is selected where the Mean Excess Function $e(u) = E[X - u | X > u]$ is most linear (low curvature), indicating the GPD assumption is valid. **GPD fit:** For exceedances $y_i = x_i - u^*$ (losses above threshold):

$$P(Y \leq y) = 1 - \left(1 + \frac{\xi y}{\sigma}\right)^{-1/\xi}$$

where ξ = shape (tail index), σ = scale. **VaR formula (POT):**

$$\text{VaR}_p = u^* + \frac{\sigma}{\xi} \left[\left(\frac{1-p}{\bar{F}(u^*)} \right)^{-\xi} - 1 \right]$$

CVaR (Expected Shortfall):

$$\text{CVaR}_p = \frac{\text{VaR}_p + \sigma - \xi u^*}{1 - \xi}$$

Fat tail warning: If $\xi > 0.5$, the variance is theoretically infinite. AlphaAgent warns: "Extremely fat tails — Black Swan risk."

Method 2: Block Maxima — GEV Distribution

Divide the return series into non-overlapping blocks of 21 trading days. Take the maximum loss in each block. Fit a Generalized Extreme Value (GEV) distribution:

$$P(M_n \leq x) = \exp \left[- \left(1 + \xi \frac{x - \mu}{\sigma} \right)^{-1/\xi} \right]$$

GEV VaR:

$$\text{VaR}_p^{\text{GEV}} = \mu + \frac{\sigma}{\xi} [(-\log p)^{-\xi} - 1]$$

GEV is used as a cross-check against the POT estimate. Divergence between the two signals model instability.

Tail Dependence Coefficient (λ_L)

Measures the probability that the stock crashes given the market (SPY) crashes simultaneously. Uses an empirical (non-parametric) estimator:

$$\hat{\lambda}_L = \frac{\#\{X_t < Q_q(X) \text{ and } B_t < Q_q(B)\}}{\#\{B_t < Q_q(B)\}}$$

where $q = 0.05$ (worst 5% of market returns). Values near 1 mean the stock always crashes with the market. AlphaAgent warns when $\lambda_L > 0.5$.

EVTResult Fields

Field	Description
<code>var_95</code>	GPD 95% Value at Risk (negative = loss)
<code>var_99</code>	GPD 99% VaR
<code>cvar_95</code>	GPD Expected Shortfall at 95%
<code>cvar_99</code>	GPD Expected Shortfall at 99%
<code>gev_var_95</code>	GEV Block Maxima 95% VaR (cross-check)
<code>tail_index</code>	GPD shape ξ : positive = fat tails
<code>tail_dependence</code>	λ_L co-crash probability with SPY [0,1]
<code>n_tail_events</code>	Exceedances used for GPD fit

Chapter 12 — Kalman Filter (Dynamic Beta)

Source: `quant_engine/kalman.py`

Theory

Static OLS regression produces a single beta coefficient over the entire historical window. But betas change over time — a stock's sensitivity to the market shifts with macroeconomic regime, sector rotation, and company-specific events. The Kalman Filter treats beta as a latent state that evolves as a random walk and updates it optimally given new price data.

State-Space Formulation

State vector: $\theta_t = [\alpha_t, \beta_t]^\top$ (intercept + slope) **Observation equation:**

$$r_{\text{stock},t} = \alpha_t + \beta_t \cdot r_{\text{SPY},t} + \epsilon_t, \quad \epsilon_t \sim N(0, R)$$

State transition (random walk):

$$\theta_t = \theta_{t-1} + \eta_t, \quad \eta_t \sim N(0, Q)$$

where $Q = \frac{\delta}{1-\delta} I_2$ controls how quickly beta is allowed to change ($\delta = 10^{-4}$ by default — slow drift).

Kalman Recursion

Predict:

$$\hat{\theta}_{t|t-1} = \theta_{t-1}, \quad P_{t|t-1} = P_{t-1} + Q$$

Innovation:

$$\nu_t = r_{\text{stock},t} - H_t \hat{\theta}_{t|t-1}, \quad H_t = [1, r_{\text{SPY},t}]$$

Kalman Gain:

$$K_t = P_{t|t-1} H_t^\top (H_t P_{t|t-1} H_t^\top + R)^{-1}$$

Update:

$$\begin{aligned} \theta_t &= \hat{\theta}_{t|t-1} + K_t \nu_t \\ P_t &= (I - K_t H_t) P_{t|t-1} \end{aligned}$$

KalmanResult Fields

Field	Description
beta	Current dynamic beta (time-varying market sensitivity)
alpha	Current dynamic alpha (idiosyncratic daily return)
spy_correlation	Rolling 60-day correlation with SPY
beta_series	Full historical time series of beta (for plotting)

Chapter 13 — Topological Data Analysis (TDA)

Source: `quant_engine/tda_signal.py` · Library: `ripser` (C++ persistent homology)

Theory

TDA applies algebraic topology to detect the "shape" of financial data that standard statistical measures miss. Persistent homology tracks how topological features (connected components = H_0 , loops = H_1) appear and disappear as a filtration scale parameter grows. Long-lived features are "persistent" and considered genuine structure.

Pipeline

Step 1 — Takens Delay Embedding: Reconstruct a high-dimensional attractor from the 1D return series using delay coordinates:

$$\mathbf{x}_i = (r_i, r_{i+\tau}, r_{i+2\tau}, \dots, r_{i+(d-1)\tau}) \in \mathbb{R}^d$$

Default: $d = 3$ (dimension), $\tau = 2$ (lag). Normalized to $[0, 1]^d$.

Step 2 — Vietoris-Rips Complex: Build a simplicial complex by connecting points within distance ϵ . Grow ϵ from 0 to max. Track birth and death of holes. **Step 3 — Persistent Homology:**

- H_0 barcodes: connected components (birth when cluster forms, death when merged)
- H_1 barcodes: loops (birth when loop forms, death when filled)
- Lifetime = death – birth

Step 4 — Signal derivation:

$$\text{TDA Signal} = \begin{cases} -h_{1,\text{norm}} & \text{if } h_1 > 0.6 \text{ or } \text{Betti}_1 \geq 3 \quad (\text{CYCLIC}) \\ 0 & \text{if entropy} > 0.7 \quad (\text{FRAGMENTED}) \\ +1 & \text{if } h_1 < 0.2 \text{ and entropy} < 0.4 \quad (\text{TRENDING}) \end{cases}$$

Regime Labels

Label	H1 max	H0 entropy	Signal	Trading Implication
TRENDING	Low	Low	+1.0	Momentum strategies work — price moves directionally
CYCLIC	High	Any	−H1	Mean-reversion strategies work — price loops around attractor
FRAGMENTED	Low	High	0.0	Regime transition — no clear directional or reverting bias
NEUTRAL	Mixed	Mixed	Interpolated	Moderate trend/cycle mix

TDAResult Fields

Field	Description
<code>h0_max_persistence</code>	Max lifetime of connected components
<code>h1_max_persistence</code>	Max lifetime of 1-cycles (loops)
<code>h0_persistence_entropy</code>	Shannon entropy of H0 diagram (fragmentation measure)
<code>betti_0</code>	H0 features at mid-scale (connected components surviving)
<code>betti_1</code>	H1 features at mid-scale (loops surviving)
<code>tda_signal</code>	[−1, +1]: +1=trending, −1=cyclic/mean-reverting
<code>regime_label</code>	TRENDING / CYCLIC / FRAGMENTED / NEUTRAL

Chapter 14 — Hawkes Self-Exciting Process

Source: `quant_engine/hawkes.py`

Theory

In a Hawkes process, each event increases the probability of further events — the process is self-exciting. In finance, a large price move (event) begets further large moves via informed trader responses, stop-loss cascades, and market maker hedging. A branching ratio approaching 1 signals that each event triggers nearly one child event — the process is explosive.

Exponential Hawkes Process

Conditional intensity:

$$\lambda(t) = \mu + \alpha \sum_{t_i < t} e^{-\beta(t-t_i)}$$

Parameters:

- μ = background rate (baseline event frequency per day)
- α = excitation magnitude (each event's triggering strength)
- β = decay rate (how fast excitement fades)
- α/β = branching ratio n (must be < 1 for stationarity)

Event definition: Returns exceeding $\pm 1.5\sigma$ (configurable) are extracted as events, converting the continuous return series to a point process $\{t_1, t_2, \dots, t_N\}$.

MLE Fitting (Ozaki 1979)

Log-likelihood:

$$\ell(\mu, \alpha, \beta) = \sum_{i=1}^N \log \lambda(t_i) - \int_0^T \lambda(t) dt$$

Integral term (closed-form via recursion):

$$\int_0^T \lambda(t) dt = \mu T + \frac{\alpha}{\beta} \sum_{i=1}^N \left(1 - e^{-\beta(T-t_i)}\right)$$

Recursive A_i (efficient computation):

$$A_i = e^{-\beta(t_i - t_{i-1})}(1 + A_{i-1}), \quad \lambda(t_i) = \mu + \alpha A_i$$

Optimizer: SLSQP with constraint $\beta > \alpha$ (guarantees branching ratio < 1 at convergence).

Cascade Risk Classification

Branching Ratio	Level	Interpretation
< 0.50	LOW	Events rarely self-propagate. Normal market.
0.50–0.80	MODERATE	Some clustering. Elevated HFT or informed trading.
0.80–0.95	HIGH	Strong clustering. Possible dark pool footprint or cascade start.
≥ 0.95	CRITICAL	Near-explosive cascade. Institutional domino risk. Alert fired.

Dark Pool Signal

$$\text{DarkPool} = \text{clip} \left(0.5 \cdot \frac{\alpha}{\beta} + 0.5 \cdot \text{clip} \left(\frac{(e/e_0 - 1)}{2}, 0, 1 \right), 0, 1 \right)$$

where e = observed event rate, $e_0 = 0.05$ (5% baseline). High clustering + above-average event rate suggests institutional block trading in dark pools.

Chapter 15 — Technical Agent (30 Factors)

Source: `agents/technical.py` · Agent name: `"technical"`

The Technical Agent analyzes price action, momentum, options flow, and market microstructure. It uses purely price- and options-derived signals — no fundamental or macro data. It is most predictive over 1–5 day horizons.

Factor Group 1 — Classical Technical Indicators

1. RSI — Relative Strength Index (14-period)

$$RSI = 100 - \frac{100}{1 + RS}, \quad RS = \frac{\text{Avg Gain (14d)}}{\text{Avg Loss (14d)}}$$

Scoring: RSI < 30 → oversold → score ~80 (bullish). RSI > 70 → overbought → score ~20 (bearish). RSI 30–70 → score ~50 (neutral). Linear interpolation between thresholds.

Parameter: `technical.rsi_window = 14` (dynamic, from settings.yaml)

2. MACD Histogram

$$MACD = EMA_{12} - EMA_{26}$$

$$\text{Signal Line} = EMA_9(MACD)$$

$$\text{Histogram} = MACD - \text{Signal}$$

Scoring: Histogram > 0 and rising → score 70–80. Histogram < 0 and falling → score 20–30. Crossover events scored 75 (bullish cross) or 25 (bearish cross).

3. Bollinger Band Width (Squeeze Signal)

$$\text{BB Width} = \frac{\text{Upper Band} - \text{Lower Band}}{\text{Middle Band}}$$

$$\text{Upper} = \text{SMA}_{20} + 2\sigma_{20}, \quad \text{Lower} = \text{SMA}_{20} - 2\sigma_{20}$$

Scoring: Narrow bands (squeeze) = compression → breakout expected → score 65.

Wide bands (expansion) = already moving → score based on position within bands.

Price above upper band → overbought → score 25.

Parameter: `technical.bollinger_window = 20`, `technical.bollinger_std = 2.0`

4. ADX — Average Directional Index

$$\text{ADX} = \text{EMA}_{14} \left(\frac{|\text{+DI} - \text{-DI}|}{\text{+DI} + \text{-DI}} \times 100 \right)$$

Scoring: $\text{ADX} > 25$ = strong trend → score 70–80 if price advancing, 30–20 if declining. $\text{ADX} < 20$ = weak/no trend → score ~50 (neutral). ADX measures trend strength, not direction.

5. Golden/Death Cross

SMA-50 vs SMA-200 crossover. If $\text{SMA-50} > \text{SMA-200}$ → Golden Cross (bullish) → score 72. If $\text{SMA-50} < \text{SMA-200}$ → Death Cross → score 28. If no crossover occurred → 55 (modest bullish/bearish depending on gap size).

6. EMA 9/21 Crossover

Fast EMA-9 vs slow EMA-21. If $\text{EMA-9} > \text{EMA-21}$ → short-term bullish → score 65. If $\text{EMA-9} < \text{EMA-21}$ → bearish → score 35. Crossover within 3 days → stronger signal ± 10 points.

7. Momentum Acceleration

$$\text{Mom} = \frac{P_t}{P_{t-22}} - 1, \quad \text{Accel} = \text{Mom}_{\text{recent}} - \text{Mom}_{\text{lagged}}$$

Scoring: Positive and accelerating momentum → score 70–85. Decelerating despite positive momentum → score drops toward 50. Negative and accelerating → score 15–30.

8. Variance Risk Premium (VRP)

$$\text{VRP} = \text{IV}_{\text{ATM}} - \sigma_{\text{realized}, 30\text{d}}$$

Positive VRP = options overpriced relative to realized vol = expected mean reversion. Score: VRP > 5% → bearish (score 35), VRP near 0 → neutral (50), VRP < -5% → options cheap, bullish (65).

Factor Group 2 — Options Microstructure

9. IV Skew (25Δ Put-Call)

$$\text{IV Skew} = \text{IV}_{\text{OTM put}} - \text{IV}_{\text{OTM call}}$$

$$\text{OTM put} : K \in [0.85 \cdot S, 0.96 \cdot S] \quad \text{OTM call} : K \in [1.04 \cdot S, 1.15 \cdot S]$$

Interpretation: Positive skew = puts more expensive than calls = fear premium = market expects downside. Score: skew > 5% → score 30 (bearish). Skew < 0 → calls expensive → bullish/complacent → score 70. Near 0 → neutral 50.

10. Gamma Exposure (GEX)

Black-Scholes Gamma:

$$\Gamma = \frac{e^{-d_1^2/2}}{S \cdot \sigma \cdot \sqrt{T} \cdot \sqrt{2\pi}}, \quad d_1 = \frac{\ln(S/K) + (r + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}$$

Net GEX (\$ per 1% move):

$$\text{GEX} = \sum_i \Gamma_i^{\text{call}} \cdot \text{OI}_i^{\text{call}} \cdot 100 \cdot S - \sum_j \Gamma_j^{\text{put}} \cdot \text{OI}_j^{\text{put}} \cdot 100 \cdot S$$

Interpretation: Positive GEX → dealers long gamma → buy dips, sell rallies → volatility suppression → score 65. Negative GEX → dealers short gamma → amplify moves → score 35. Extreme negative GEX → potential pin risk → score 20.

11. Max Pain

$$\text{MaxPain} = \arg \min_K \left[\sum_i \max(K - K_i, 0) \cdot \text{OI}_i^{\text{call}} + \sum_j \max(K_j - K, 0) \cdot \text{OI}_j^{\text{put}} \right]$$

Interpretation: Max pain is the strike where option holders lose the most money at expiration — market makers are said to push price toward this level. Gap = (MaxPain – Spot)/Spot × 100. Gap > 3% up → bullish (score 68). Gap > 3% down → bearish (score 32). Neutral within ±1% (score 50).

12. Implied Correlation (vs SPY)

$$\text{Impl. Corr.} = \frac{IV_{\text{SPY,ATM}}}{IV_{\text{Stock,ATM}}}$$

Interpretation: Ratio > 1 = stock has lower implied vol than market → stock likely to move less than market → score 60. Ratio < 0.5 = stock implied vol much higher than market → idiosyncratic risk premium → score 35. Note: this is a proxy for the formal implied correlation basket formula.

Factor Group 3 — Signal Quality

13–14. TDA (Topological Data Analysis) Signal

See Chapter 13. The Technical Agent calls `TDASignalEngine().analyze(prices)` and adds the TDA signal as a factor. Score: `50 + tda_signal × 30` (range: 20 to 80). TRENDING regime → score 80. CYCLIC → score 20.

15–19. Additional Indicators (Stochastic, ATR, Trend Slope, Volume, CMF)

Stochastic Oscillator (%K/%D), ATR-normalized trend, 20-day volume z-score, Chaikin Money Flow, price position within 52-week range. All scored on the same 0–100 bullish/bearish scale.

30. PCA Signal Quality

Normalizes all factor scores to $[-1, +1]$, computes mean signal direction, average signal strength, dispersion (std), and consensus as `strength × max(0, 1 - dispersion/1.5)`. Score: `50 + consensus × 30`. Low dispersion + strong direction → high quality signal (score 70–80). High dispersion + weak direction → poor quality signal (score 45–55).

Composite Probability Calculation

```
raw_scores = [f.score for f in factor_scores.values()]
avg_score = mean(raw_scores)          # 0-100
prob_up = avg_score / 100.0           # 0-1
# Clamp to [0.10, 0.90] to avoid extreme posteriors
prob_up = max(0.10, min(0.90, prob_up))
vote = LONG if prob_up > 0.55 else SHORT if prob_up < 0.45 else HOLD
```

Chapter 16 — Fundamental Agent (28 Factors)

Source: `agents/fundamental.py` · Agent name: `"fundamental"`

The Fundamental Agent assesses company quality, valuation, and financial health. It draws on `yfinance` financials, FRED risk-free rates, and Fama-French factor theory. It is most predictive over 3-month to 1-year horizons.

Factor Group 1 — Valuation Multiples

1. P/E Ratio (Trailing)

Fetches from `info["trailingPE"]`. Score: P/E < 15 → value territory → score 75. P/E 15–25 → fair value → score 55. P/E 25–40 → slightly expensive → score 40. P/E > 40 → growth premium or speculative → score 25. P/E thresholds:

`fundamental.pe_cheap=15`, `pe_fair=25`, `pe_expensive=40`.

2. PEG Ratio

$$\text{PEG} = \frac{\text{P/E}}{\text{EPS Growth Rate (\%)}}$$

PEG < 1 → undervalued for growth → score 78. PEG 1–2 → fair → score 55. PEG > 2 → expensive → score 30. Thresholds: `peg_undervalued=1.0`, `peg_fair=2.0`.

3. EV/EBITDA

< 10 → cheap → score 80. 10–20 → fair → score 55. 20–30 → elevated → score 40. > 30 → expensive → score 20. Thresholds: `ev_ebitda_cheap=10`, `ev_ebitda_fair=20`, `ev_ebitda_expensive=30`.

4. Price/Book Ratio

< 1.5 → value territory → score 75. 1.5–3.0 → fair → score 50. > 3.0 → growth premium → score 35. Thresholds: `pb_undervalued=1.5`, `pb_fair=3.0`.

5. Price/Sales Ratio

< 2 → cheap → score 75. 2–4 → fair → score 55. > 4 → expensive → score 35. Thresholds: `ps_cheap=2.0`, `ps_fair=4.0`.

6. FCF Yield

$$\text{FCF Yield} = \frac{\text{Free Cash Flow}}{\text{Market Cap}} \times 100$$

FCF Yield > 5% → strong → score 80. 2–5% → adequate → score 60. < 2% → weak → score 35. Thresholds: `fcf_yield_good=2.0`, `fcf_yield_great=5.0`.

Factor Group 2 — DCF Intrinsic Value

7. DCF Upside

$$V = \sum_{t=1}^5 \frac{\text{FCF}_t}{(1+r)^t} + \frac{\text{FCF}_5 \cdot (1+g)}{(r-g)(1+r)^5}$$

where r = WACC (estimated), g = terminal growth = 3%.

$$\text{DCF Upside} = \frac{V - P}{P} \times 100\%$$

Upside > 30% → strongly undervalued → score 85. Upside 10–30% → moderately undervalued → score 70. –20 to 10% → near fair value → score 50. Below –20% → overvalued → score 25.

Factor Group 3 — Quality Metrics

8–10. Revenue Growth, Earnings Growth

Year-over-year growth from income statements. Revenue growth > 20% YoY → score 80. 10–20% → score 65. < 10% → score 45. Negative growth → score 25.

11. ROE — Return on Equity

$$\text{ROE} = \frac{\text{Net Income}}{\text{Shareholders' Equity}}$$

> 20% → score 80. 10–20% → score 60. < 10% → score 40. Negative → score 20.

Thresholds: `roe_strong=20`, `roe_moderate=10`.

12. ROA — Return on Assets

> 15% → score 80. 8–15% → score 60. 3–8% → score 45. < 3% → score 25.

`roa_strong=15`, `roa_moderate=8`, `roa_weak=3`.

13. Operating Margin

> 25% → score 80. 15–25% → score 65. 5–15% → score 50. < 5% → score 25.

`operating_margin_strong=25`, `operating_margin_moderate=15`.

14. Altman Z-Score

$$Z = 1.2X_1 + 1.4X_2 + 3.3X_3 + 0.6X_4 + 1.0X_5$$

where X_1 = Working Capital/Assets, X_2 = Retained Earnings/Assets, X_3 = EBIT/Assets, X_4 = Market Cap/Liabilities, X_5 = Revenue/Assets.

$Z > 3.0$ → safe → score 80. Z 2.0–3.0 → grey zone → score 55. $Z < 2.0$ → distress risk → score 25. Threshold: `fundamental.altman_z_quality=3.0`.

Factor Group 4 — Advanced Theories

15–19. Fama-French 5-Factor Model

$$R_i - R_f = \alpha + \beta_m(R_m - R_f) + \beta_s \text{SMB} + \beta_v \text{HML} + \beta_r \text{RMW} + \beta_c \text{CMA} + \epsilon$$

- **SMB** (Small Minus Big): Size premium. Positive loading = small-cap tilt → higher expected return.
- **HML** (High Minus Low P/B): Value premium. High P/B → growth stock → lower expected return historically.
- **RMW** (Robust Minus Weak): Profitability premium. High operating profitability → higher expected return.
- **CMA** (Conservative Minus Aggressive): Investment premium. Conservative investment → higher expected return.

Factor proxies (yfinance): SMB from market cap vs median, HML from P/B ratio, RMW from operating margin, CMA from capex ratio.

20. CAPM Jensen's Alpha

$$\text{Expected Return} = R_f + \beta(R_m - R_f)$$

$$\alpha = R_i^{1\text{yr}} - \text{Expected Return}$$

where R_f = FRED FEDFUNDS rate (annualized decimal), R_m = SPY 1-year return, β = 1-year OLS beta of stock vs SPY. **Scoring:** α in percentage $\times 1.5 + 50$. Alpha +10% → score 65. Alpha -10% → score 35. Clamped to [10, 90].

21. Dividend Growth Rate (5yr CAGR)

$$\text{DGR} = \left(\frac{D_n}{D_0} \right)^{1/(n-1)} - 1$$

where D_n, D_0 = most recent / oldest annual dividend, n = years of history.

DGR > 10% → strong growth → score 75. DGR > 5% → moderate → score 62. DGR > 0% → positive → score 55. No dividend or negative growth → score 45. From

```
yf.Ticker(ticker).dividends.resample("YE").sum()
```

22. Lockup Expiration Proxy

Uses `info["firstTradeDateMilliseconds"]` to estimate IPO date. Days since IPO: if 150–200 days (lockup typically expiring) → warning issued, score 40 (bearish: potential insider selling). 0–150 days (pre-lockup) → score 48. > 200 days → lockup passed → score 55.

23. Accruals Ratio

$$\text{Accruals Ratio} = \frac{\text{Net Income} - \text{Operating Cash Flow}}{\text{Total Assets}}$$

High accruals = earnings quality is poor (income not backed by cash). Accruals < -5% → high quality earnings → score 75. Accruals 0–5% → moderate → score 55. Accruals > 10% → red flag → score 25.

24. Forward P/E

From `info["forwardPE"]`. Lower forward P/E than trailing P/E = earnings expected to grow = positive → score 70. Higher forward P/E = earnings expected to fall = bearish → score 35.

25. Net Margin Trend

Compares current net margin vs prior year. Expanding margin → score 72.

Contracting margin → score 32. Stable (within 1%) → score 52.

26. Buyback Signal

If shares outstanding fell YoY → company is buying back stock (return of capital) → bullish signal → score 70. Shares rising → dilution → score 35. Stable → score 52.

27–28. EPS Surprise + Interest Coverage

EPS surprise: actual vs estimated %. Coverage ratio: EBIT/Interest expense. Strong coverage ($>10\times$) → score 78. Adequate ($5-10\times$) → score 60. Weak ($<2\times$) → score 25.

Chapter 17 — Macro Agent (26 Factors)

Source: `agents/macro.py` · Agent name: `"macro"`

Factor Group 1 — Monetary Policy

1. Fed Funds Rate

From FRED `FEDFUNDS`. Rate < 2% → easy money → bullish equities → score 70. Rate 2–4% → neutral → score 55. Rate > 4% → restrictive → score 35.

`macro.fed_rate_easy=2.0`, `fed_rate_tight=4.0`.

2. Yield Curve (10Y – 2Y)

Spread = FRED `GS10` – FRED `GS2`. Inverted yield curve (spread < 0) historically precedes recession by 12–18 months.

Spread > 0.5% → healthy curve → score 70. Spread –0.5% to 0.5% → flat/flattening → score 50. Spread < –0.5% → inverted → recession warning → score 25.

`macro.yield_curve_positive=0.5`, `yield_curve_inverted=-0.5`.

3. CPI Inflation (YoY)

From FRED `CPIAUCSL`. CPI < 2.5% → comfortable → score 70. 2.5–5% → elevated but manageable → score 50. > 5% → high inflation erodes real returns → score 30.

`macro.cpi_target=2.5`, `cpi_elevated=5.0`.

4. M2 Money Supply Growth

From FRED `M2SL`. YoY M2 growth > 5% → accommodative monetary conditions → bullish → score 65. M2 shrinking → quantitative tightening → score 35.

5. SOFR Spread (Repo Market Stress)

$$\text{SOFR Spread} = \text{SOFR} - \text{Fed Funds Rate}$$

SOFR = Secured Overnight Financing Rate (FRED series `SOFR`). Normal: spread near 0. Spread > 0.10% → mild stress → score 70. Spread > 0.25% → elevated repo stress → score 50. Spread > 0.50% → crisis-level → score 30 + warning. Repo market stress often precedes broader financial stress.

6. Fed Balance Sheet (WALCL)

From FRED `WALCL` (Assets: Total Assets: Total Assets — quantitative easing gauge). Growing balance sheet → accommodative → score 68. Shrinking (QT) → tightening → score 38.

Factor Group 2 — Market Breadth & Risk

7. VIX Fear Index

From yfinance `^VIX`. VIX < 20 → calm → score 70. VIX 20–30 → elevated concern → score 50. VIX > 30 → fear regime, but often contrarian buy signal near peaks → score 35 (bearish near term but 50 medium term).

8. Credit Spreads (HYG/LQD)

Computes the 1-month return spread between HYG (high yield) and LQD (investment grade). Widening spreads (HYG underperforms) = credit stress = bearish → score 30. Tightening spreads = risk appetite = bullish → score 70.

9. Copper/Gold Ratio

$$\text{Cu/Au Ratio} = \frac{P_{\text{Copper}}}{P_{\text{Gold}}}$$

Copper = industrial demand proxy (bullish growth). Gold = fear/store of value. Rising ratio → economic growth → score 72. Falling ratio → growth fears → score 32.

10. Global Equity Breadth (ACWI vs SPY)

1-month relative return of ACWI (global) vs SPY (US). Positive = global outperforming US = broader risk-on appetite = bullish → score 65. Negative = global weakness despite US strength = fragile rally → score 40.

11. SOX vs SPY (Semiconductor Relative Strength)

SOXX (semiconductor ETF) vs SPY 1-month relative return. Semis lead tech cycles → outperforming semis → tech bullish signal → score 68. Semis lagging → cycle top caution → score 38.

12. Global Pre-Market Signal (Asia/EU)

Average overnight return of EWJ (Japan), FXI (China), EFA (Europe) ETFs. Positive = risk-on globally → US open likely bullish → score 65. Negative = risk-off → score 38.

13. Growth vs Value Rotation (IVW/IVE)

1-month relative return of growth ETF IVW vs value ETF IVE. Growth outperforming = risk-on, momentum market → score 62. Value outperforming = defensive rotation → score 45.

Factor Group 3 — Activity Indicators

14. ISM Manufacturing PMI

From FRED `MANEMP` or proxy. PMI > 55 → strong expansion → score 78. PMI 50–55 → modest expansion → score 62. PMI < 50 → contraction → score 32.

`pmi_expansion=50`, `pmi_strong=55`, `pmi_weak=48`.

15. Initial Jobless Claims

From FRED `IC4WSA`. Claims < 220,000 → tight labor market → score 75. 220–260K → moderate → score 55. > 350K → labor market stress → score 25.

`macro.claims_low=220000`, `claims_elevated=260000`.

16. UMich Consumer Sentiment

From FRED `UMCSENT`. > 90 → strong → score 75. 70–90 → moderate → score 58. < 55 → weak → score 30. `consumer_sentiment_strong=90`.

17. TIPS Breakeven Inflation

From FRED `T10YIE` (10-year breakeven). 1.5–2.5% → goldilocks range → score 68. > 3% → high inflation expectations → score 30. < 1.5% → deflationary fears → score 42.

`tips_breakeven_low=1.5`, `tips_breakeven_high=2.5`.

18. Recession Probability

Internal estimate based on yield curve inversion + unemployment + PMI sub-50. Risk < 20% → score 72. 20–60% → score 45. > 60% → high recession risk → score 22.

`macro.recession_prob_low=0.2`, `recession_prob_cutoff=0.6`.

19. BTC Risk-On Signal

3-month BTC return. BTC > +20% → strong risk appetite → bullish equities → score 68. BTC < -20% → risk-off → score 35. `macro.btc_momentum_bull=20`,

`btc_momentum_bear=-20`.

20. Amihud Illiquidity Ratio

$$\text{Amihud} = \frac{1}{T} \sum_{t=1}^T \frac{|r_t|}{V_t}$$

where r_t = daily return, V_t = daily dollar volume. Applied to SPY as a market-wide liquidity gauge.

Low Amihud (liquid market) → score 65. High Amihud (illiquid, large moves per dollar) → score 35.

21. Bond-Equity Correlation (TLT/SPY)

60-day rolling correlation of TLT (20Y Treasury) vs SPY returns. Negative correlation → traditional diversification working → stable risk parity environment → score 62. Positive correlation (bonds and stocks falling together) → liquidity crisis signal → score 35.

26. PCA Signal Quality

Same PCA quality methodology as Technical Agent (see Chapter 15). Applied to all macro factor scores to measure internal consensus.

Chapter 18 — Sentiment & Behavioral Agent

14 factors | Ticker: `sentiment` | RAG pipeline + behavioral finance quantification

Architecture Overview

The Sentiment Agent is the only agent that calls an LLM at inference time. It builds a Retrieval-Augmented Generation (RAG) pipeline: news headlines are fetched via NewsAPI, stored in a ChromaDB vector store, and the top-5 most semantically relevant articles for the current ticker are retrieved using cosine similarity. The retrieved context is passed to Gemini 2.0 Flash with a structured prompt asking for a SCORE (0–100) and two-sentence REASON. All remaining factors are quantitative and run unconditionally.

Keyword fallback: If Gemini is unavailable, `_keyword_scan()` counts bullish words ("surge", "beat", "growth", "upgrade") vs bearish words ("drop", "miss", "loss", "downgrade") and returns 0.65, 0.35, or 0.50.

Factor 1 — News Sentiment (RAG)

RAG Pipeline

Query: `"lawsuits, earnings, bankruptcy, growth, new products, {ticker}"`. Time filter: last 12 hours only. Gemini returns 0–100 integer → divided by 100 → `rag_prob`. Probability update: `prob_up += (rag_prob - 0.5) × 0.4`. Confidence set to 0.8 if context contains "lawsuit", "earnings", "bankrupt", "fda", "sec"; else 0.55.

Factor 2 — Short Interest

Calculation

From `yfinance` `info`: `shortPercentOfFloat` ($\times 100$) and `shortRatio` (days to cover).
 Score: `clamp(80 - short_pct × 2, 10, 90)`. Threshold
`agent_defaults.short_interest_high = 20%`: above → `prob_up -= 0.08`, squeeze warning. 10–20%: `prob_up -= 0.03`. Below 3%: `prob_up += 0.03`.

Factor 3 — Fear & Greed Index

Contrarian Logic

Score ≤ 25 (Extreme Fear) $\rightarrow$ contrarian buy $\rightarrow$ `prob_up += 0.05`. Score ≥ 75 (Extreme Greed) $\rightarrow$ contrarian caution $\rightarrow$ `prob_up -= 0.05`. Computed internally from VIX, breadth, put/call metrics via AlternativeData module.

Factor 4 — Analyst Consensus

Formula

$$\text{bull_pct} = \frac{\text{StrongBuy} + \text{Buy}}{\text{Total Analysts}}$$

`bull_pct > 0.70` $\rightarrow$ `prob_up += 0.05`. `bear_pct > 0.30` $\rightarrow$ `prob_up -= 0.04`.

Threshold: `agent_defaults.analyst_bullish_consensus = 0.70`.

Factor 5 — Market Breadth (RSP vs SPY)

RSP (equal-weight S&P 500) vs SPY (cap-weight). RSP outperforming $\rightarrow$ broad advance $\rightarrow$ score >65 . SPY only leading $\rightarrow$ narrow mega-cap rally $\rightarrow$ score <35 . Score >65 : `prob_up += 0.03`. Score <35 : `prob_up -= 0.03`.

Factor 6 — Options Put/Call Skew

Near-term options chain: `P/C skew = total_put_volume / total_call_volume`. <0.7 $\rightarrow$ bullish speculation $\rightarrow$ score 70. >1.2 $\rightarrow$ heavy put hedging $\rightarrow$ score 25. Neutral $\rightarrow$ score 50.

Factor 7 — News / Social Momentum

Count of yfinance news articles in last 3 days. Score: $\min(80, 40 + \text{recent_count} \times 4)$.
High news volume = social attention spike, precedes above-average short-term volatility.

Factor 8 — EPS Forward Revision

$$\text{revision}\% = \frac{\text{forwardEPS} - \text{trailingEPS}}{|\text{trailingEPS}|} \times 100$$

>15% → score 85. 5–15% → score 60. –5 to +5% → score 50. <–5% → score 25.

Factor 9 — Unusual Options Activity (UOA)

$\text{vol_oi_ratio} = \text{call_volume} / \text{call_open_interest}$. Score: $\min(85, 40 + \text{vol_oi_ratio} \times 30)$. Ratio >1.5 = fresh positioning, potential informed buying.

Factor 10 — Transfer Entropy Proxy (News→Price)

$$\hat{T}E_{\text{news} \rightarrow \text{price}} = \text{Corr}(\text{news_count}_{t-1}, \text{return}_t)$$

Granger-causality proxy. $|\text{corr}| > 0.3 \rightarrow$ "news reliably leads price". Score: $\text{clamp}(50 + \text{corr} \times 40, 10, 90)$.

Factor 11 — Shannon Entropy (Signal Clarity)

$$H = - \sum_i p_i \ln p_i \quad H_{\text{norm}} = H / \ln N \quad \text{score} = (1 - H_{\text{norm}}) \times 80 + 10$$

Low entropy = all factors aligned (high conviction). High entropy = maximum internal disagreement.

Factor 12 — Analyst Price Target Upside

$$\text{upside} \% = \frac{\text{targetMeanPrice} - \text{currentPrice}}{\text{currentPrice}} \times 100$$

>20% → score 85. >10%: `prob_up += 0.04`. <-10%: `prob_up -= 0.04`.

Factor 13 — AAI Sentiment Survey (Contrarian)

FRED series `AAIIBULL` and `AAIIBEAR`. Bull-bear spread <-20% → extreme bearishness → contrarian buy → score 75, `prob_up += 0.04`. Spread >30% → extreme bullishness → contrarian caution → score 25, `prob_up -= 0.03`.

Factor 14 — Consumer Credit Proxy (Margin Debt)

FRED `TOTALSL` (total consumer credit). 3-month rate of change: rising credit = leveraged retail demand. Score 60 if >2%, 50 if positive, 40 if contracting.

Chapter 19 — Insider & Whale Agent

11 factors | Ticker: `insider` | EDGAR Form 4, Kyle's Lambda, cluster buying, institutional positioning

Architecture Overview

The Insider Agent detects "smart money" activity via SEC EDGAR Form 4 filings and institutional ownership data. Unlike other agents, final probability is derived as a *mean of all factor scores* (equal weighting, not additive update):

$$\text{prob_up} = \text{sigmoid-map} \left(\frac{1}{N} \sum_i \text{score}_i, [0, 100] \right)$$

Factor 1 — Insider Sentiment (Net Transaction Volume)

From yfinance `insider_transactions`: `net_shares =  $\Sigma$ buys -  $\Sigma$ sells`. Score maps net shares to 0–100 centered at 50. Confidence += 0.2 when `|net_shares| > 50,000`.

Factor 2 — Institutional Backing (Whale Sentiment)

From yfinance `major_holders`: institutional ownership %. >80% → high backing → score ~78. <30% → score ~35. Confidence += 0.2 when >80%.

Factor 3 — EDGAR Form 4 Activity Count

Form 4 filings (last 60 days) from SEC EDGAR. ≥ 5 → high cluster → score 65 (insiders buying ≥ 5 reports in 60 days is a statistically rare event). 0 → score 50. 1–4 → score 55.

Factor 4 — Material Events (8-K Count)

8-K filings in last 30 days. $\geq 5 \rightarrow$ elevated material event risk $\rightarrow$ score 30, warning added. $1-4 \rightarrow$ score 50. Earnings calls, executive changes, asset sales, and legal proceedings are all filed as 8-Ks.

Factor 5 — EDGAR Institutional Ownership (Cross-Check)

13F-based ownership from EDGAR. Score: $\min(80, \text{edgar_inst_pct} \times 0.8)$. Cross-validates yfinance data with official SEC filings.

Factor 6 — Insider Cluster Buys

From yfinance `insider_purchases`: count buy transactions. Score: $\min(85, 40 + \text{buy_count} \times 12)$. ≥ 3 transactions = "strong cluster" — multiple insiders buying simultaneously is one of the most reliable bullish microstructure signals.

Factor 7 — Institutional Ownership % (13F)

$>70\% \rightarrow$ score 80. $40-70\% \rightarrow$ score 60. $<40\% \rightarrow$ score 40. From yfinance `institutional_holders`.

Factor 8 — Short Squeeze Potential

$$\text{squeeze_score} = \min(85, 40 + \text{short_float} \% \times 1.2)$$

$>15\%$ short float $\rightarrow$ "squeeze risk" warning. High short + rising price = self-reinforcing buying loop.

Factor 9 — Float Reduction (Buyback vs Dilution)

Quarterly balance sheet: `share_chg = (shares_now - shares_old) / shares_old × 100`.
`<-2% → active buyback → score 85. >+3% → dilution warning, score 20.`

Factor 10 — Kyle's Lambda (Market Microstructure)

Theory (Kyle 1985)

$$\Delta P_t = \lambda \cdot Q_t^{\text{signed}} + \varepsilon_t \quad \hat{\lambda} = \frac{\text{Cov}(\Delta P, Q^{\text{signed}})}{\text{Var}(Q^{\text{signed}})}$$

λ = price impact per unit of signed order flow. High λ = thin market (large moves per share). Low λ = liquid (institutions can hide orders). Normalized: `lambda_norm =  $|\lambda| \times 10^6 / \text{avg\_price}$` . Score: `clamp(70 - lambda_norm × 10, 10, 90)`.

Factor 11 — Insider Cluster 30-Day (EDGAR)

EDGAR Form 4 filings filed within the last 30 days. ≥ 3 filings in 30 days → strong recent cluster → score 75. More timely than the 60-day count in Factor 3.

Chapter 20 — Risk & Circuit Breaker Agent

16 factors | Ticker: `risk` | 5 hard override circuits + continuous risk metrics

Override Priority Cascade

Priority	Override	Trigger	Effect
1	BLACK SWAN	Z-score in last 5 sessions $> 5\sigma$	prob_up=0.10, multiplier=0.0 (halt)
2	FLASH CRASH	Ticker 1d $< -7\%$ OR SPY 1d $< -5\%$	prob_up=0.15, multiplier=0.0 (halt)
3	GEO SHOCK	VIX > 35 AND (Gold $> +2\%$ OR Oil $> +5\%$)	prob_up=0.30, multiplier=0.35
4	CARRY UNWIND	USD/JPY < 125 OR JPY $+1.5\%$ in 1d	prob_up=0.30, multiplier=0.5, SHORT
5	GARCH EXTREME	vol_regime=EXTREME OR VaR99 $< -5\%$	prob_up=0.20, multiplier=0.0
6	GARCH HIGH	vol_regime=HIGH OR VaR95 $< -3\%$	prob_up=0.40, multiplier=0.5

Factor 1 — GARCH Volatility Regime

GARCHModel fits GARCH(1,1) on 1-year returns, forecasts 5-day forward vol. Regime set by percentile vs historical distribution: $< 25\text{th pctl} \rightarrow \text{LOW (score 75)}$. $25\text{--}75\text{th} \rightarrow \text{NORMAL (50)}$. $75\text{--}95\text{th} \rightarrow \text{HIGH (25)}$. $> 95\text{th} \rightarrow \text{EXTREME (5)}$.

Factor 2 — EVT Tail Risk (99% VaR)

GPD fit to worst 10% of returns (see Chapter 11). VaR99 $> -2\% \rightarrow \text{score 80}$. $-2\text{ to }-5\% \rightarrow \text{score 40}$. $< -5\% \rightarrow \text{score 10 (extreme tails)}$.

Factor 3 — Monte Carlo 95% CI (5-Day)

5000-path GBM using GARCH-forecasted vol. CI95.low > -3% → score 75. -3 to -7% → score 40. < -7% → score 10.

Factor 4 — Kelly Position Size

$$f^* = \frac{p \cdot b - q}{b}, \quad b = \frac{\text{win_return}}{\text{loss_return}}, \quad q = 1 - p$$

Half-Kelly: $f^*/2$. Vol-adjusted Kelly: $f^*/2 \times (\text{target_vol} / \text{current_vol})$. Cap: 25%.

Inputs: `prob_win` = fraction of MC paths above current price (converted from percentage). `win_return` = (CI68.high – price) / price. `loss_return` = |CI68.low – price| / price.

Factors 5–8 — Black Swan, Flash Crash, Geo Shock, Carry Unwind

Black Swan: Z-score of recent returns $>5\sigma$ → under a normal distribution this is a ~1-in-3.5M daily event → halt all trades.

Flash Crash: Ticker drops >7% in a single session OR SPY drops >5% → halt all trades.

Geo Shock: Requires ALL of: VIX >35 AND (GLD 1d >+2% OR XLE 1d >+5%). Conjunction prevents single-variable false positives. Cap position at 35%.

Carry Unwind: USD/JPY <125 → yen surging → carry trade forced liquidation → 50% size, SHORT bias.

Factor 9 — KL Divergence (Regime Shift)

$$D_{KL}(P_{recent} || P_{baseline}) = \sum_{i=1}^{20} p_i \ln \frac{p_i}{q_i}$$

Bins: 20 equal-width bins over [min, max] of baseline returns. Laplace smooth: +1e-10.

KL < 0.4 → stable → score ~70. KL 0.4–1.0 → drift → score ~40. KL > 1.0 → regime break → score 10, critical warning.

Factor 10 — Hawkes Branching Ratio

Fit Hawkes process (Chapter 14) to return jumps. Branching ratio α/β : >0.9 → near-critical cascade risk. Score: `clamp(70 - branching_ratio × 60, 10, 90)`.

Factor 11 — Quasi-MC VaR (Sobol)

4096-path Sobol GBM simulation (Chapter 10). 5-day 95th percentile worst case. Same scoring as Factor 3. ~10× sample efficiency vs pseudorandom MC.

Factor 12 — Liquidity Risk (Volume Ratio)

`liq_ratio = last_vol / avg_20d_vol`. >1.5 → liquid → score 80. <0.7 → thin → score 30. Parameters: `risk.liquidity_ratio_high = 1.5`, `risk.liquidity_ratio_low = 0.7`.

Factor 13 — Drawdown from All-Time High

$$DD\% = \frac{\text{last_close} - \text{ATH}_{2yr}}{\text{ATH}_{2yr}} \times 100 \quad \text{score} = \text{clamp}(90 + DD\% \times 1.5, 10, 90)$$

>30% below ATH → warning added. Parameter: `risk.drawdown_warning_pct = 30`.

Factor 14 — Tail Ratio (Up/Down Return Asymmetry)

$$\text{tail_ratio} = P_{95}/|P_5|$$

>1.2 → positive skew → score 80. 0.8–1.2 → symmetric → score 50. <0.8 → fat downside tails → score 25.

Factor 15 — Correlation Regime (vs SPY)

Rolling 60d corr vs SPY minus 252d corr. Large negative Δ (<-0.1) → decorrelating → idiosyncratic risk → score 70. Large positive Δ → market-driven → score 30.

Threshold: `risk.correlation_delta_threshold = 0.1`.

Factor 16 — GARCH Standard Circuit Breakers

Steady-state (no emergency override): EXTREME or $\text{VaR}_{99} < -5\%$ → $\text{prob_up} = 0.20$, halt. HIGH or $\text{VaR}_{95} < -3\%$ → $\text{prob_up} = 0.40$, half-size. Normal → $\text{prob_up} = 0.50$, full size.

Chapter 21 — Geopolitical Agent

19 factors | Ticker: `geopolitical` | ETF-based geo-risk proxies, FRED GPR Index

Architecture Overview

All signals are derived from liquid ETF prices and FRED data — no specialized geopolitical API needed. Geopolitical events manifest in ETF prices within hours. The agent monitors XLE (oil), GLD (gold), ITA (defense), EEM (emerging markets), HYG/LQD (credit), ACWI (global equity), and the Caldara-Iacoviello GPR Index from FRED.

Factor 1 — Oil Shock (XLE vs SPY)

```
_rel_strength("XLE", "SPY", lookback=22): 22-day RS. >+5% → energy stress →
prob_up -= 0.08. <-5% → calm → prob_up += 0.03. Score: clamp(50 - RS×2, 10,
90).
```

Factor 2 — Gold Safe-Haven (GLD)

```
GLD 22-day momentum. >+5% → safe-haven flight → prob_up -= 0.10, conf+=0.05.
<-3% → easing → prob_up += 0.05. Score: clamp(50 - momentum×3, 10, 90).
```

Factor 3 — EM Stress (EEM vs SPY)

```
EEM RS vs SPY (22d). <-5% → EM stress → prob_up -= 0.06. >+3% → healthy →
prob_up += 0.04. Score: clamp(50 + RS×2.5, 10, 90).
```

Factor 4 — Defense RS (ITA vs SPY)

```
ITA RS >+5% → defense stocks bid up → conflict risk priced → prob_up -= 0.07.
Defense outperformance often leads geopolitical news by weeks.
```

Factor 5 — VIX Fear Index

$>30 \rightarrow \text{extreme fear} \rightarrow \text{prob_up} -= 0.10$, warning. $20-30 \rightarrow \text{elevated} \rightarrow \text{prob_up} -= 0.05$. $<20 \rightarrow \text{calm} \rightarrow \text{prob_up} += 0.03$. Score: $\text{clamp}(90 - (\text{VIX}-10) \times 2.5, 10, 90)$.

Factor 6 — Copper/Gold Ratio

COPX/GLD ratio 1-month change. Rising $\rightarrow$ industrial demand expanding $\rightarrow$ growth signal. $>+3\% \rightarrow \text{prob_up} += 0.05$. $<-3\% \rightarrow$ demand contraction $\rightarrow \text{prob_up} -= 0.05$.

Factor 7 — Global Breadth (ACWI vs SPY)

ACWI RS $<-3\% \rightarrow$ global equities lagging US $\rightarrow$ international headwinds $\rightarrow \text{prob_up} -= 0.04$. $>+2\% \rightarrow$ healthy $\rightarrow \text{prob_up} += 0.03$.

Factor 8 — USD Safe-Haven (DXY)

From CurrencyData. DXY 1m $>+2\% \rightarrow$ safe-haven USD buying $\rightarrow$ risk-off $\rightarrow \text{prob_up} -= 0.06$. $<-2\% \rightarrow$ risk-on $\rightarrow \text{prob_up} += 0.04$.

Factor 9 — Credit Stress (HYG/LQD)

HYG/LQD ratio 1-month change. Ratio falling $\rightarrow$ credit spreads widening $\rightarrow \text{prob_up} -= 0.07$. Rising $\rightarrow$ credit healthy $\rightarrow \text{prob_up} += 0.04$. Credit stress typically leads equity stress by weeks.

Factor 10 — Sector Rotation Signal

$$\text{rotation_score} = 1.5 \times RS_{XLK} - 1.0 \times RS_{XLE} - 0.5 \times RS_{XLF}$$

Tech leading → growth regime → bullish. Energy/financials leading → inflation/stagflation. $>+5$: `prob_up += 0.03`. <-5 : `prob_up -= 0.04`.

Factor 11 — Supply Chain Stress (Shipping BDRY)

BDRY (Breakwave Dry Bulk Shipping ETF) 1-month momentum. Proxy for Baltic Dry Index. Collapse $<-20\%$ → demand destruction warning. Score: `clamp(50 + bdry_mom×1.5, 10, 90)`.

Factor 12 — Commodity Shock (XLE + XLB)

Average 1-month momentum of XLE (energy) + XLB (materials). $>+8\%$ → commodity inflation. Score: `clamp(50 + avg_mom×1.2, 10, 90)`.

Factor 13 — EM Contagion Risk (EMB)

EMB (J.P. Morgan EM Bond ETF) 1-month momentum. $<-3\%$ → capital flight from EM → leading indicator of global risk-off. Score: `clamp(50 + emb_mom×5, 10, 90)`.

Factor 14 — Oil Price Direct (Brent BZ=F)

Brent crude 22-day momentum. $>+15\%$ → inflationary tension → `prob_up -= 0.05`. $<-15\%$ → demand concern → `prob_up += 0.03`. Score: `clamp(50 - oil_mom×1.5, 10, 90)`.

Factor 15 — Election Cycle Phase

Days to next US election (2026 midterms Nov 3, 2028 presidential Nov 7). <90d → policy uncertainty → score 40. 90–365d → pre-election drift → score 55. >365d → early cycle → score 60.

Factor 16 — Transport Index (IYT vs SPY — Dow Theory)

Dow Theory: bull market requires both Industrials and Transports at new highs. IYT RS >+3% → transports confirming → score 65. IYT RS <−3% → divergence → score 35.

Factor 17 — Commodity Index (PDBC)

PDBC (Invesco Commodity ETF) 22-day momentum. >+12% → commodity inflation → `prob_up -= 0.04`. Score: `clamp(50 + crb_mom, 10, 90)`.

Factor 18 — GPR Index (FRED Geopolitical Risk)

Caldara-Iacoviello GPR Index

Published by the Federal Reserve (FRED: `GPRC` or `GPRGLOBAL`). Based on automated text search of newspaper archives for geopolitical risk keywords (war, terrorism, political tensions).

$$\text{GPR_rel} = \frac{\text{GPR_now}}{\text{GPR_historical_mean}}$$

$$\text{Score} = \text{clamp}(70 - (\text{GPR_rel} - 1.0) \times 40, 10, 90)$$

>2.0 → extreme risk → `prob_up -= 0.08`, critical warning. 1.5–2.0 → elevated → `prob_up -= 0.04`. <0.75 → calm tailwind → `prob_up += 0.03`.

Factor 19 — PCA Signal Quality

Applied to all Geopolitical factor scores. Measures internal consensus: $\text{consensus} = \text{signal_strength} \times \max(0, 1 - \text{dispersion}/1.5)$. Score: $\text{clamp}(50 + \text{mean_signal} \times 40, 10, 90)$.

Chapter 22 — Volatility Agent

9 factors | Ticker: `volatility` | Options surface, GARCH regime, Kalman beta, VIX term structure

Architecture Overview

The Volatility Agent reads the options market and GARCH-derived volatility to assess whether current conditions are bullish or bearish. High-vol environments accompany falling prices; low-vol environments accompany upward drift. This agent also provides the dynamic Kalman beta used by Bayesian Fusion for correlation adjustment.

Factor 1 — GARCH Vol Regime

LOW → `prob_up += 0.10`, `conf+=0.10`. HIGH → `prob_up -= 0.15`, `conf+=0.15`.
EXTREME → `prob_up -= 0.20`, `conf+=0.20`. Score map: {LOW:70, NORMAL:50, HIGH:30, EXTREME:10}.

Factor 2 — Put/Call Ratio

Total put OI / total call OI. >1.2 → heavy hedging → `prob_up -= 0.15`. <0.7 → bullish speculation → `prob_up += 0.10`. Parameters: `volatility.put_call_overbought = 1.2`, `volatility.put_call_bearish = 0.7`.

Factor 3 — IV vs Realized Vol (Variance Risk Premium)

$$\text{IV_RV_ratio} = \frac{\text{ATM Implied Move}}{\sigma_{\text{GARCH_daily}} \times \sqrt{T_{\text{trading}}}}$$

>1.5 → options overpriced, vol crush likely → slight bullish → `prob_up += 0.05`. <0.7 → options cheap, vol expansion expected → caution. Parameters:

`volatility.iv_rv_overpriced = 1.5`, `volatility.iv_rv_underpriced = 0.7`.

Factor 4 — Kalman Dynamic Beta

Time-varying beta from state-space Kalman Filter (see Chapter 12). Score=50 (directionally neutral — beta quantifies magnitude of market exposure, not direction). Used by Bayesian Fusion for correlation-adjusted weighting.

Factor 5 — SPY Correlation (60-Day Rolling)

Rolling 60-day Pearson correlation vs SPY. Score=50 (neutral for direction). Feeds into Bayesian Fusion correlation penalty: high correlation → less independent signals per agent → lower incremental weight.

Factor 6 — VVIX (Volatility of Volatility)

^VVIX measures implied volatility of VIX options. >120 (`volatility.vvix_extreme`) → options market pricing in major VIX spike. Score: `clamp(100 - (VVIX-60)×1.5, 10, 90)`.

Factor 7 — VIX Term Structure (Contango/Backwardation)

VIX 3M (^VIX3M) – spot VIX. Positive → contango, calm → score 75. Negative → backwardation, acute panic → score 25. <-2 → "acute market stress" warning in reasoning.

Factor 8 — Realized Vol Trend (10d/30d)

$$RV_N = \sigma_{Nd} \times \sqrt{252} \times 100\% \quad \text{ratio} = RV_{10}/RV_{30}$$

<0.8 → vol calming → score 70. >1.2 → vol spiking → score 25. Parameters:

`volatility.rv_short_window = 10`, `volatility.rv_long_window = 30`.

Factor 9 — CBOE SKEW Index (Tail Risk Premium)

^SKEW measures the price of S&P 500 OTM puts relative to ATM options. High SKEW while VIX is low = "fear of the unknown" — black swan protection being purchased quietly. >140 → high tail risk → score <30. Score: `clamp(130 - SKEW, 10, 90)`.

Chapter 23 — Currency Agent

11 factors | Ticker: `currency` | FX dynamics, carry trade, sector revenue exposure weighting

Architecture Overview

FX impact is stock-specific: a stronger USD hurts multinationals more than domestic companies. `SECTOR_FX_EXPOSURE` maps each sector to an estimated international revenue percentage. The DXY change is multiplied by this fraction and a 0.3 transmission coefficient to estimate the per-stock earnings headwind/tailwind.

Factor 1 — US Dollar Index (DXY)

$$\text{FX_impact} = -\Delta \text{DXY}_{1m} \times \text{sector_exposure} \times 0.3$$

Tech (55% intl rev) + DXY up 3%: impact = -0.495 → prob_up reduced by 0.005.

Score: `clamp(50 - dxy_1m×4, 10, 90)`. Multiplier: `currency.fx_impact_multiplier = 0.3`.

Factor 2 — EUR/USD Rate

>1.05 → strong euro → USD weak → favorable for US multinationals → score 65.

<0.95 → headwind → score 35. EUR/USD is ~57% of the DXY index weighting.

Factor 3 — USD/JPY Carry Signal

Borrow in JPY (near-zero rates), invest in USD assets. When USD/JPY falls sharply, carry traders sell risk assets to repay JPY. <120 → carry unwind → `prob_up -= 0.08`, warning. >150 → carry intact → slight bullish. Parameters: `currency.usdjpy_weak = 120`, `currency.usdjpy_strong = 140`.

Factor 4 — USD/CNY (China Stress)

>7.3 → CNY weakening → China macro stress → score 30, `prob_up -= 0.04`.

Parameter: `currency.usdcny_stress = 7.3`.

Factor 5 — GBP/USD Rate

>1.25 → strong pound → score 65. <1.25 → score 40. Relevant for UK-exposed companies.

Factor 6 — EM Currency Stress

Average USD gain vs EM currency basket. >+3% → capital flight → `prob_up -= 0.06`, warning. <-2% → risk appetite → `prob_up += 0.04`. Score: `clamp(50 - em_stress×5, 10, 90)`.

Factor 7 — FX Translation Impact

$$\text{translation} = -\Delta DXY_{1m} \times \text{sector_exposure} \times 0.25$$

Accounting translation of foreign earnings into USD. A DXY +4% with 50% foreign revenue → ~2% less earnings in USD terms.

Factor 8 — Petro-Currency Signal (CAD/AUD)

Average 22-day momentum of CAD=X and AUD=X. Rising petro-currencies → commodity demand strong → risk-on. >+2% → `prob_up += 0.03`. <-2% → `prob_up -= 0.03`.

Factor 9 — Carry Trade Attractiveness (Rate Differential)

FRED `FEDFUNDS` minus 3.5% global benchmark proxy. Rate diff $>+1\%$ → USD carry attractive → capital inflows → score 70. Narrowing carry → score 35.

Factor 10 — Real Interest Rate (10Y)

$$\text{Real Rate} = \text{DGS10} - \text{T10YIE}$$

Positive real rate → USD attractive, gold headwind. $>+1\%$ → score 65. 0 to $+1\%$ → score 50. $<0\%$ → score <40 (negative real rates erode USD appeal).

Factor 11 — EM FX Pressure (EEM Proxy)

EEM 22-day momentum. $>+5\%$ → EM strength → score ~ 60 . $<-5\%$ → capital flight → score ~ 40 . Score: `clamp(50 + eem_mom*2, 10, 90)`.

Chapter 24 — Risk Circuit Breakers & Override System

24.1 Two-Layer Override Architecture

AlphaAgent has two override layers: (1) Agent-level overrides inside the Risk Agent that modify its own output via `prob_up` and `multiplier` fields. (2) Orchestrator-level response to the `multiplier` field when computing the final SignalPacket position size.

24.2 Override Keyword Detection

Keyword in reasoning	System Response	Position Size
BLACK SWAN	Halt all new positions	0%
FLASH CRASH	Halt all new positions	0%
WAR/GEO SHOCK	Emergency size cap	35%
CARRY UNWIND	SHORT bias enforced	50%
CRITICAL RISK	Capital protection mode	0%
HIGH RISK	Half-position mode	50%

24.3 Position Size Formula

$$\text{final_size} = \text{kelly_fraction} \times \text{risk_multiplier} \times \text{portfolio_value}$$

24.4 Geopolitical Override

The Geopolitical Agent independently adds: if `avg_score < 25` → "GEOPOLITICAL OVERRIDE: Extreme risk — cap position size at 35%". This can trigger simultaneously with a Risk Agent override, producing compounding size reductions.

Chapter 25 — Dynamic Horizon Weighting

25.1 Motivation

RSI has predictive power over days; DCF value materializes over years. One fixed set of agent weights cannot serve all horizons. The horizon system reblends agent outputs post-orchestration.

25.2 Horizon Weight Table

Horizon	Technical	Vol	Sentiment	Macro	Fundamental	Insider	Currency	Geo	Risk
1D	3.0	3.0	2.0	0.5	0.2	0.5	0.5	0.3	1.0
1W	2.5	2.5	1.8	0.8	0.3	0.8	0.8	0.5	1.0
1M	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
3M	0.5	0.5	0.7	1.5	2.0	2.0	1.2	1.5	1.0
6M	0.3	0.3	0.5	2.0	2.5	2.5	1.5	2.0	1.0
1Y	0.2	0.2	0.3	2.5	3.0	3.0	2.0	2.5	1.0

25.3 Implementation

$$P_{\text{horizon_raw}} = \frac{\sum_i w_i \cdot p_i}{\sum_i w_i} \quad P_{\text{final}} = 0.8 \cdot P_{\text{horizon_raw}} + 0.2 \cdot P_{\text{base}}$$

The 20% anchoring to the base probability prevents extreme reweighting from completely inverting the Bayesian fusion output. Direction re-derived: $>0.55 \rightarrow \text{LONG}$, $<0.45 \rightarrow \text{SHORT}$, else NEUTRAL. API response includes both `probability` (horizon-weighted) and `base_probability` (raw orchestrator output).

Chapter 26 — Signal Decay & Holding Period (Ornstein-Uhlenbeck)

26.1 Holding Period Estimation

AlphaAgent estimates holding period using the Ornstein-Uhlenbeck mean-reversion half-life. If log-price follows an OU process, the half-life gives the expected time for a price deviation to revert 50% back to the mean — approximating the optimal holding period before alpha decays.

$$dX_t = \theta(\mu - X_t) dt + \sigma dW_t \quad t_{1/2} = \frac{\ln 2}{\theta}$$

The API response includes `holding_period.min_days`, `expected_days`, and `max_days` (68% confidence interval of the OU process).

Chapter 27 — Settings Configuration Reference

Key Parameters by Section (config/settings.yaml)

data

Key	Default	Purpose
technical_period	1y	Technical indicator lookback
short_period	5d	VIX term structure, flash crash
medium_period	3mo	Liquidity ratio rolling average
long_period	2y	ATH, correlation regime
min_rolling_days	20	Minimum for rolling averages
min_tail_days	60	Minimum returns for tail ratio
backtest_period	3y	Default historical backtest window
walk_forward_train_years	1.0	Walk-forward training window (years)

technical

Key	Default	Purpose
rsi_window	14	RSI Wilder smoothing period
bollinger_window	20	Bollinger Bands SMA period
bollinger_std	2.0	Bollinger Bands standard deviation
adx_window	14	ADX trend strength period
momentum_lookback	22	1-month price momentum (trading days)
min_data_days	50	Minimum price history to run agent

volatility

Key	Default	Purpose
regime_low_pct	25	Percentile → LOW vol regime
regime_high_pct	95	Percentile → HIGH vol regime
put_call_bearish	0.7	P/C below → bullish
put_call_overbought	1.2	P/C above → bearish hedging
iv_rv_overpriced	1.5	IV/RV → vol crush likely
vvix_extreme	120	VVIX → extreme meta-uncertainty
skew_extreme	140	SKEW → high tail risk
rv_short_window	10	Short RV window (days)
rv_long_window	30	Long RV window (days)

fundamental

Key	Default	Purpose
pe_cheap / pe_fair / pe_expensive	15 / 25 / 40	P/E ratio thresholds
peg_undervalued / peg_fair	1.0 / 2.0	PEG ratio thresholds
fcf_yield_good / fcf_yield_great	2% / 5%	Free cash flow yield thresholds
dcf_upside_strong / dcf_upside_moderate	30% / 10%	DCF upside % for scoring
revenue_growth_strong	20%	YoY revenue growth → strong score
roe_strong / roe_moderate	20% / 10%	Return on equity thresholds
altman_z_quality	3.0	Altman Z → financially safe

macro

Key	Default	Purpose
yield_curve_positive	0.5%	10Y-2Y → normal slope
yield_curve_inverted	-0.5%	10Y-2Y → inversion (recession)
fed_rate_easy / fed_rate_tight	2.0% / 4.0%	Fed Funds policy thresholds
cpi_target / cpi_elevated	2.5% / 5.0%	CPI threshold levels
pmi_expansion / pmi_strong	50 / 55	ISM PMI regime levels
claims_low / claims_elevated	220K / 260K	Initial jobless claims levels

risk

Key	Default	Purpose
liquidity_ratio_high / low	1.5 / 0.7	Volume ratio thresholds
drawdown_warning_pct	30%	ATH drawdown → warning
tail_ratio_positive / negative	1.2 / 0.8	95th/5th pctl asymmetry
correlation_delta_threshold	0.1	60d vs 252d corr Δ → regime change

backtest

Key	Default	Purpose
initial_capital	\$100,000	Starting portfolio value
max_position_pct	20%	Max allocation per position
transaction_cost_bps	5 bps	Commission: 0.05% one-way
bid_ask_spread_bps	2 bps	Market impact slippage
borrow_cost_annual	0.50%	Short selling borrow cost
kelly_fraction	10%	Kelly fraction for position sizing

agent_defaults

Key	Default	Purpose
neutral_probability	0.5	All agents start here
long_threshold	0.55	prob_up above → LONG vote
short_threshold	0.45	prob_up below → SHORT vote
rag_high_confidence	0.8	Confidence when material keywords detected

Chapter 28 — API Reference

28.1 Core Signal Endpoint

```
GET /api/v1/signal/{ticker}?horizon=1m
```

Runs full 9-agent LangGraph analysis. Horizon: `1d`, `1w`, `1m`, `3m`, `6m`, `1y` (default: `1m`).

```
{
  "ticker": "AAPL",      "horizon": "3m",
  "direction": "LONG",   "probability": 0.6312,
  "base_probability": 0.5891, "conviction": 74.2,
  "multiplier": 0.85,    "entropy": 0.312,
  "agents": [{ "name": "technical", "vote": "LONG",
                "probability_up": 0.72, "confidence": 0.85,
                "reasoning": "...", "factor_scores": {...} }],
  "warnings": [], "holding_period": {"min_days":3,"expected_days":12,"max_days":28},
  "summary": {"bull_agents":6,"bear_agents":2,"neutral_agents":1},
  "latency_ms": 42300
}
```

28.2 All Endpoints

Method	Endpoint	Description
GET	/	Serves dashboard HTML
GET	/api/v1/signal/{ticker}?horizon=	Full 9-agent signal
GET	/api/v1/portfolio	Live portfolio positions
POST	/api/v1/paper/signal/{ticker}	Signal + paper trade
POST	/api/v1/backtest	Historical backtest
POST	/api/v1/chat	LLM chat about signal
GET	/api/v1/quant/garch/{ticker}	GARCH forecast only
GET	/api/v1/quant/monte-carlo/{ticker}	Monte Carlo only
GET	/api/v1/quant/evt/{ticker}	EVT tail risk only
GET	/api/v1/chart/history/{ticker}	OHLCV bars (5m/1h/1d)
GET	/api/v1/markets	US + global index prices
POST	/api/v1/portfolio/add	Add position to DB
POST	/api/v1/portfolio/close/{ticker}	Close position in DB
GET	/api/v1/leaderboard	Agent accuracy rankings
POST	/api/v1/optimizer	Portfolio weight optimization
POST	/api/v1/stress-test	Scenario stress test
WS	/ws/{ticker}	Real-time agent reasoning stream

Chapter 29 — Dashboard & Frontend Architecture

29.1 Single-File Architecture

The entire dashboard is `frontend/index.html` (~1800 lines). All HTML, CSS, and JavaScript in one file — no build step, no npm. FastAPI serves it from `/`. Only external dependency: Lightweight Charts (TradingView) from CDN.

29.2 Dashboard Tabs

Tab	Key Features	Primary API
Signal Analysis	9-agent deep research, 1D–1Y horizon selector, Top-50 company chips, smart autocomplete, TradingView chart (1D/5D/1M/3M/6M/1Y), per-agent factor detail cards	/api/v1/signal/{ticker}?horizon=
Live Portfolio	Real-time positions, unrealized P&L, portfolio composition chart, add/close positions	/api/v1/portfolio
Paper Trading	Simulated order execution, trade log, running P&L	/api/v1/paper/signal/{ticker}
Backtest	Equity curve, Sharpe, max drawdown, win rate, trade log	/api/v1/backtest
Walk-Forward	Rolling OOS validation, Sharpe stability chart	/api/v1/backtest (walk-forward mode)
Stress Test	2008 crash, COVID-19, 1987, rate shock, oil shock, tech bubble scenarios	/api/v1/stress-test
Quant Lab	Interactive GARCH, Monte Carlo visualization, EVT, Sobol QMC	/api/v1/quant/*
Leaderboard	Agent accuracy ranking, confidence calibration, recent call history	/api/v1/leaderboard
Optimizer	Mean-variance optimization, efficient frontier	/api/v1/optimizer

29.3 Key JavaScript Patterns

Helper: `function $(id){ return document.getElementById(id); }`

API wrapper: `async function api(path, opts)` wraps fetch with error handling; relative paths (no hardcoded host).

Chart periods: `CHART_PERIODS` array drives button rendering. `1d` → 5-minute bars (yfinance period="1d" interval="5m"). `5d` → hourly bars.

Company autocomplete: `COMPANY_DB` (51 entries). `onTickerInput()` — prefix match on ticker first, then substring match on company name, shows 8 results. Arrow-key navigation + Enter to select.

Horizon buttons: `sigHorizon` JS variable. `runSignal()` appends `?`
`horizon=${sigHorizon}`. Active button gets class `hz-active`.

Chapter 30 — Backtesting, Walk-Forward & Stress Testing

30.1 Backtest Engine (backtest/engine.py)

Replays AlphaAgent signals on historical price data with realistic execution simulation.

Parameter	Default	Description
<code>initial_capital</code>	\$100,000	Starting portfolio cash
<code>max_position_pct</code>	20%	Max allocation per position
<code>transaction_cost_bps</code>	5 bps	Commission (0.05% one-way)
<code>bid_ask_spread_bps</code>	2 bps	Slippage
<code>borrow_cost_annual</code>	0.50%	Short selling borrow cost
<code>kelly_fraction</code>	10%	Kelly fraction for sizing
<code>default_hold_days</code>	5	Default exit horizon

Backtest Flow

1. Download OHLCV for all tickers in backtest universe for the period.
2. For each date: run orchestrator on data available up to that date only (no lookahead).
3. Position sizing: $\text{kelly_fraction} \times |\text{prob_up} - 0.5| \times 2 \times \text{max_position_pct}$.
4. Cost model: buy at `price × (1 + (transaction_cost + bid_ask)/10000)`. Sell symmetric. Short: deduct daily borrow.
5. Exit: after `default_hold_days` or signal direction flip.
6. Compute Sharpe, Sortino, max drawdown, win rate, profit factor, CAGR.

$$\text{Sharpe} = \frac{R_p - R_f}{\sigma_p} \sqrt{252} \quad \text{Sortino} = \frac{R_p - R_f}{\sigma_{\text{down}}} \sqrt{252}$$

$$\text{Max Drawdown} = \max_t \frac{\text{Peak}_{[0,t]} - \text{Trough}_{[t,T]}}{\text{Peak}_{[0,t]}} \quad \text{Profit Factor} = \frac{\sum \text{wins}}{|\sum \text{losses}|}$$

30.2 Walk-Forward Validation (backtest/walk_forward.py)

Rolling windows prevent in-sample overfitting. Training period: 1 year (252 trading days). Test period: 3 months (63 trading days). Step: 1 month.

```
|—— Train (252d) ——|— Test (63d) —|
      |—— Train (252d) ——|— Test (63d) —|
            |—— Train (252d) ——|— Test (63d) —|
```

Key metrics: OOS Sharpe, OOS win rate, Sharpe stability (std across windows). OOS win rate < 50% → strategy is likely noise.

30.3 Stress Testing (backtest/stress_test.py)

Scenario	Key Shocks
2008 Financial Crisis	S&P -55%, VIX → 80, credit spreads +500bps
COVID-19 Crash	S&P -34% in 33 days, VIX → 85, oil -70%
1987 Black Monday	Single-day -22.6%
Taper Tantrum (2013)	10Y yield +1.4% in 3 months
Rate Shock (+200bps)	10Y +2% sudden, tech P/E compression
Oil Shock	Oil +80%, inflation +3%
Tech Bubble Burst	Nasdaq -78% over 2 years

30.4 Paper Trading (trading/paper_trader.py)

Simulated brokerage in SQLite. Calls signal API → executes at current market price.

Tracks open positions, realized P&L, unrealized P&L, trade log, daily equity curve. Primary validation step before real capital deployment.

30.5 RL Portfolio Rebalancer (trading/rl_rebalancer.py)

PPO-based reinforcement learning agent for portfolio rebalancing. State: current weights + recent returns + GARCH vol regime. Action: weight adjustment vector Δw . Reward:

Sharpe of next N-day returns. Parameters: `rl.gamma = 0.99`, `rl.learning_rate = 0.0003`.

Enhancement layer on top of the signal system — not a replacement.

Epilogue — System Summary

AlphaAgent converges three intellectual traditions:

1. **Classical Quantitative Finance:** GARCH(1,1), Bayesian inference, Kelly Criterion, EVT, Kalman filtering — methods developed 1950s–1990s that remain foundational.
2. **Modern Machine Learning:** Hidden Markov Models, Topological Data Analysis, Hawkes processes, Quasi-Monte Carlo, Reinforcement Learning.
3. **AI-Native Architecture:** LangGraph multi-agent parallel orchestration, Retrieval-Augmented Generation for news sentiment, Gemini 2.0 Flash for real-time LLM inference.

163 factors across 9 specialist agents → Correlation-adjusted Bayesian fusion → Time-horizon reweighting → 6-tier risk circuit breakers → Final LONG/SHORT/NEUTRAL signal with conviction score, Kelly position size, and OU-estimated holding period. Typical latency: 30–60 seconds per signal.

Disclaimer: AlphaAgent is an educational system for research and learning. It is NOT financial advice, NOT a registered investment advisor, and NOT intended for live trading without extensive validation, regulatory compliance, and risk management oversight. Past backtest performance does not guarantee future results.

AlphaAgent Technical Reference — v2.0 — Phase 6 Complete — May 2026